

EDUCATION

- **BTech, Electrical Engineering**, National Institute of Technology, Jaipur 2025

EXPERIENCE

ICICI Bank — Data Engineer, Mumbai July 2025 – Present

- Worked with IBM InfoSphere DataStage and Designer to schedule and monitor automated ETL jobs, ensuring reliable batch processing and timely data availability across enterprise systems.
- Analyzed and resolved UCIC (Unique Customer Identification Code) mismatches by investigating MDM deduplication rules and API matching logic, improving customer identity accuracy across downstream applications.
- Built and enhanced Azure ETL pipelines using Azure Data Factory and Azure Databricks to ingest and transform source-system data, delivering analytics-ready datasets to business and risk teams.
- Implemented PII and PCI data redaction using regular expressions to identify and obscure sensitive fields across on-premise Oracle and MySQL databases, strengthening data security and compliance controls.

PROJECTS

Credit Default Risk Prediction 🔗

- Built a supervised credit default prediction model on customer financial and behavioural data, performing EDA and feature engineering such as income-to-loan and EMI-to-income ratios to better capture repayment capacity.
- Modelled non-linear risk patterns using an artificial neural network instead of linear models, improving separation between defaulters and non-defaulters.
- Evaluated model performance using ROC-AUC to address class imbalance and focus on risk ranking rather than raw accuracy, achieving an ROC-AUC of 0.87 on test data.

Financial Text Summarization & Insight Generation (NLP + LLM) 🔗

- Analysed and implemented a text summarisation pipeline for financial reports using transformer-based models, achieving a ROUGE-L score of 0.72.
- Improved summary coherence through preprocessing and chunking strategies for long documents by implementing a RAG pipeline.
- Reduced average document length by 60% while preserving key risk insights for stakeholder consumption.

Customer Spending & Risk Analytics 🔗

- Performed end-to-end customer spending analysis by querying transactional data from PostgreSQL and integrating results into Python workflows.
- Used advanced SQL constructs such as CTEs, window functions, and aggregations to identify trends, outliers, and high-risk customer segments.
- Translated analytical outputs into interpretable insights to support business and risk decision-making.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, PySpark
- **AI & ML:** PyTorch, TensorFlow, RNN, LSTM, CNN, NLP, Random Forest, XGBoost
- **Database & Analytics Tools:** ETL (IBM DataStage), MDM, Azure Data Factory, Databricks
- **Data Visualisation:** Tableau, Power BI
- **Software:** Visual Studio Code, GitHub Desktop, MS Office (Excel, Word, PowerPoint)
- **Others:** Dynatrace, Postman, Spark SQL, SQL Developer

LEADERSHIP

- **Executive, Think India MNIT Club:** Led the planning and execution of large-scale events with 5,000+ attendees, managing and coordinating a team of 20+ students to ensure smooth operations and delivery.
- **Founder, Unicorn Club (2022):** Built and scaled a community of entrepreneurs, organizing and hosting 30+ talks focused on practical business insights, collaboration, and knowledge sharing.